MOF-Derived Transition-Metal Carbide@Polypyrrole Nanohybrids for a Highly Sensitive and Stretchable Multimodal Sweat-Sensing Patch

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MOF 유래 전이 금속 탄화물@폴리피롤 나노하이브리드를 이용한 고감도 및 신축성 다중 모드 땀 감지 패치

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Abstract

Despite significant progress in wearable and stretchable electrochemical multimodal sweat sensors based on metal–organic framework (MOF)–derived nonporous carbon (NPC), sensing performance remains limited by insufficient electrochemical activity of the transduction layer. MOF-derived transition-metal carbides and their nanohybrids, featuring low charge-transfer resistance, large electrochemical surface area, and intrinsic catalytic activity, represent a highly promising class of transduction materials. Here, we report a stretchable multimodal biosensing patch based on NENU-5-derived molybdenum carbide@polypyrrole (MoC_x@PPy) nanohybrid integrated onto laser-engraved graphene(LEG) for the detection of sweat vitamin C, glucose, sodium ions, and pH. The resulting electrodes enable uniform platinum nanoparticle deposition and efficient enzyme immobilization, achieving high sensitivities of 120.98 and 154.65μA/mMcm² for vitamin C and glucose, with low limits of detection of 0.773 and 2.73μM, respectively. Near-Nernstian responses of 59.10mVdecade^{-1} for sodium ISE and −58.55mV pH^{-1}(pH 3–9) are also obtained. The multimodal patch enables reliable on-body sweat analysis during active cycling.

Keywords: ***Mo_xC@PPy nanohybrids, wearable and stretchable multimodal patch, highly sensitive Vitamin C sensor.***

1. Introduction

Wearable sweat sensors have emerged as a powerful approach to overcome the limitations of centralized and symptom-driven healthcare by enabling continuous, noninvasive, and personalized health monitoring without reliance on costly equipment or trained medical professionals.[1] Human sweat is particularly suitable for wearable sensing due to its noninvasive collection, continuous availability, and rich biochemical composition. It contains physiologically informative biomarkers, including metabolites such as glucose, lactate, uric acid, creatinine, vitamin C, dopamine, and urea, as well as electrolytes such as sodium, potassium, and calcium. Glucose monitoring is central to diabetes management, as persistently elevated levels promote oxidative stress and associated cardiovascular complications, while vitamin C and sodium provide complementary insight into antioxidant status and electrolyte balance. Recently, MOF-derived nano-porous carbon materials, transition metal carbides, and their nanohybrids have attracted increasing interest as electrochemical transduction materials owing to their high surface area, tunable composition, and enhanced charge transport properties. In this work, we present a multimodal biosensing patch based on a novel NENU-5-derived molybdenum carbide@polypyrrole (MoCx@PPy) nanohybrid for the detection of glucose, vitamin C, sodium ions, and pH in sweat. The nanohybrid is synthesized via MOF pyrolysis followed by in situ polypyrrole polymerization and integrated onto laser-engraved graphene transferred to a styrene–ethylene–butylene–styrene (SEBS) substrate. The resulting platform exhibits enhanced electrochemical activity and enables reliable on-body multimodal sweat analysis, highlighting its great potential for next-generation noninvasive personalized healthcare monitoring.

2. Results and discussion

Fig.1 shows the schematic representations of the proposed microfluidic-integrated multimodal sweat patch. The synthesis route of the NENU-5-derived molybdenum carbide@polypyrrole (MoCx@PPy) nanohybrid is illustrated in Fig. 2. The FESEM of the Mo_xC@PPy nanohybrid and the t-LEG/ MoCx@PPy/PtNPs electrode are examined by FESEM, as shown in Fig. 3. The electrochemical performance of the glucose and vitamin C sensors, evaluated through amperometric measurements, is presented in Fig. 4, whereas the open-circuit potential (OCP) responses of the sodium ion-selective electrode and the pH sensor are displayed in Fig. 5. Finally, the in vivo performance of the fully integrated multimodal biosensing patch during on-body testing is demonstrated in Fig. 6.

3. Conclusion

In this work, a microfluidic-integrated multimodal sweat sensing patch was proposed and developed using a NENU-5-derived molybdenum carbide@polypyrrole (Mo_xC@PPy) nanohybrid synthesized via MOF carbonization followed by in situ polypyrrole polymerization. The developed platform exhibited excellent sensing performance, with the vitamin C sensor demonstrating particularly high sensitivity and a low detection limit. These findings establish a generalizable materials strategy for leveraging MOF-derived transition-metal carbides and their nanohybrids in next-generation wearable biosensing platforms.

Acknowledgments

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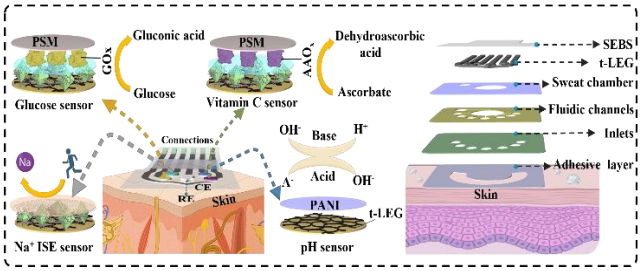

Fig. 1. Schematic illustration of the microfluidic-integrated multimodal biosensing patch.

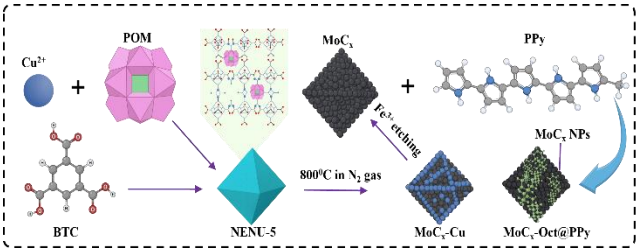

Fig. 2. Graphical overview of the synthesis route for Mo_xC@PPy nanohybrid.

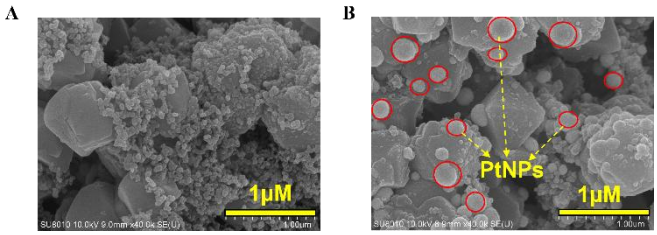

Fig. 3. (A) Field-emission scanning electron microscopy (FESEM) image of the MoCx@PPy nanohybrid, and (B) FESEM image of the t-LEG/ MoCx@PPy/PtNPs electrode.

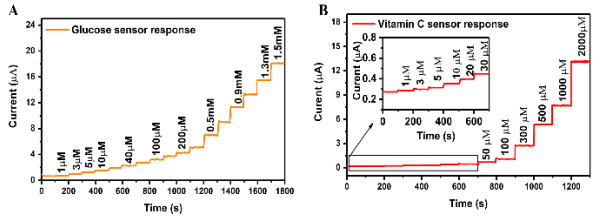

Fig. 4. (A) The amperometric response of the glucose sensor, and (B) The amperometric response of the Vitamin C sensor within their physiological range in human sweat. All the measurements were performed in 0.1 M phosphate-buffered saline (PBS, pH 7.4).

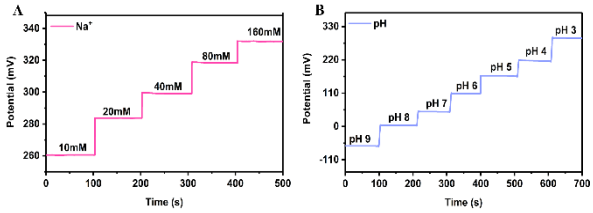

Fig. 5. (A) Open-circuit potential (OCP) response of the Na^+ ion-selective electrode. (B) OCP responses of the pH sensor in buffer solutions with pH values ranging from 3 to 9, including repeatability tests.

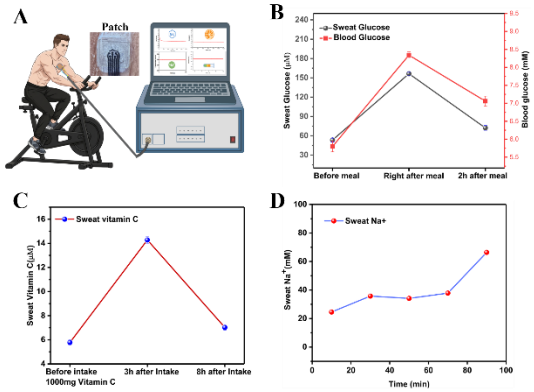

Fig. 6. (A) Schematic of the on-body configuration with an inset optical image of the multimodal patch. (B) Comparison of sweat glucose levels with blood glucose measured using a commercial glucometer. (C) Real-time amperometric response of the vitamin C sensor before and after 1000 mg vitamin C intake. (D) Time-dependent sweat sodium (Na^+) concentrations were measured using the fabricated patch with three sodium sensors.

Table 1. Comparison with other Vitamin C sensors.

Electrode	Sensitivity	Linear Range (μM)	LOD (μM)	Ref
Rh–C electrode /AAOx	N/A	0-1000	N/A	2
Au/PEDOT: $LiClO_4$ /AAOx	2.0 nA μm−1	5-5000	≈4	3
t-LEG/ Mo_xC@PPy/PtNPs/AAOx	120.98 μA/mMcm²	1-2000 1-100	≈0.77	